# Real-Valued Surrogates for the Imaginary Unit via Logarithmic Mapping Magnitude Matching

Pearl B. Pulickal

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**Abstract**

Complex numbers, particularly the imaginary unit $i$, are fundamental to analyzing linear time-invariant (LTI) systems, quantum mechanics, and signal processing. However, implementing complex-valued parameters in hardware or optimizing complex functions with real-valued algorithms can be challenging. This paper introduces a principled method for deriving a positive real scalar $x^*$ to serve as a surrogate for $i$. Using the complex-to-real mapping $f(z) = \ln z/z$, we propose selecting $x^*$ such that the "magnitude signature" $|f(x)|$ best matches $|f(i)|$.

We formulate this as the optimization problem:

$$x^* = \arg\min_{x>0} \left| \left|\frac{\ln x}{x}\right| - \left|\frac{\ln i}{i}\right| \right|.$$

Using the principal branch of the natural logarithm, $\ln i = i\pi/2$, we have $\left|\frac{\ln i}{i}\right| = \left|\frac{i\pi/2}{i}\right| = \frac{\pi}{2}$. The optimization thus becomes minimizing $||\ln x/x| - \pi/2|$ over $x > 0$. Since the maximum value of $\ln x/x$ for $x > 0$ is $1/e \approx 0.368$, which is less than $\pi/2 \approx 1.57$, the minimum is achieved when $|\ln x/x| = \pi/2$. This requires either $\ln x/x = \pi/2$ (no solution for $x > 0$) or $\ln x/x = -\pi/2$. The latter yields a unique positive solution.

We show that the equation $\ln x/x = -\pi/2$ can be solved exactly using the Lambert $W$ function, yielding the surrogate

$$x^* = \frac{2}{\pi} W\left(\frac{\pi}{2}\right) \approx 0.474541.$$

At this value, $|\ln x^*/x^*| = |-\pi/2| = \pi/2$, perfectly matching $|f(i)|$ according to the chosen criterion. We clarify that while the magnitude is matched, the value $f(x^*) = -\pi/2$ does not match $f(i) = \pi/2$. The utility of $x^*$ stems from its ability to preserve certain system characteristics despite this difference. We demonstrate this practical fidelity via a control-theoretic case study, showing that replacing $i\omega$ with $x^*\omega$ (or a structure scaled by $x^*$) in a first-order system transfer function preserves key step response and frequency response metrics to within 5%. Finally, we survey connections to realification techniques in numerical linear algebra and signal processing, highlighting the context and potential applications of this surrogate.

## 1 Introduction and Motivation

The imaginary unit $i = \sqrt{-1}$ is an indispensable tool across science and engineering, particularly for representing oscillations, rotations, and phase relationships. In linear time-invariant (LTI) systems analysis, the Laplace variable $s$ and its frequency-domain counterpart $j\omega$ (or $i\omega$) are central to characterizing system dynamics and stability. Complex numbers allow for compact representation of sinusoidal signals and elegant manipulation of impedance, transfer functions, and modal analysis.

However, translating complex-valued designs or models into real-world implementations can present significant challenges. Physical hardware often operates solely on real signals and parameters. Many standard optimization algorithms and numerical solvers are designed for real-valued inputs and outputs, and complex extensions can increase computational cost or complexity. Furthermore, certain theoretical analyses or hardware platforms (e.g., FPGAs, analog circuits) benefit from or even require real-only parameterizations.

This work addresses the challenge of finding a positive real scalar $x$ that can serve as a meaningful surrogate for the imaginary unit $i$. Instead of ad-hoc replacements, we seek a principled derivation based on matching a characteristic property derived from a complex-to-real mapping. We choose the mapping $f(z) = \ln z/z$, where ln denotes the principal branch of the natural logarithm. This function was selected due to its relevance in frequency domain analysis, linking the logarithmic phase ($\ln z$) to the frequency variable itself ($z$), capturing aspects of phase-per-frequency-unit or gain-phase trade-offs.

Our key idea is to find $x^*$ such that the "magnitude signature" of the real surrogate under this mapping, $|f(x^*)| = |\ln x^*/x^*|$, best matches the magnitude signature of the imaginary unit, $|f(i)| = |\ln i/i|$.

The contributions of this paper are:

1. To **formulate** the problem of finding a real surrogate $x^*$ for $i$ as an optimization problem minimizing the difference between $|f(x)|$ and $|f(i)|$ for $f(z) = \ln z/z$.
2. To **calculate** $|f(i)|$ using the principal branch of the logarithm.
3. To **derive** the optimal $x^*$ in closed form using the Lambert $W$ function and prove its uniqueness as the positive solution based on the chosen criterion.
4. To **clarify** the fidelity of the surrogate: precisely what property it matches (magnitude of $f$) and what properties it does not match (value and phase of $f$).
5. To **demonstrate** the practical utility and fidelity of the surrogate $x^*$ via a control-theoretic case study comparing the step and frequency responses of complex and real-surrogate systems.
6. To **contextualize** this approach within broader realification techniques used in numerical linear algebra and signal processing.

# 2 Problem Formulation and Analytic Solution

We aim to find a positive real number $x^* > 0$ that best replaces the imaginary unit $i$ under the mapping $f(z) = \ln z/z$. We define "best match" by minimizing the absolute difference between the magnitudes of the function applied to $x$ and $i$:

$$x^* = \arg\min_{x>0} \, ||f(x)| - |f(i)|| = \arg\min_{x>0} \left| \left|\frac{\ln x}{x}\right| - \left|\frac{\ln i}{i}\right| \right|.$$

First, let's evaluate $|f(i)|$. Using the principal branch of the natural logarithm, $\ln i = i\pi/2$. Thus,

$$f(i) = \frac{\ln i}{i} = \frac{i\pi/2}{i} = \frac{\pi}{2}.$$

The magnitude is therefore $|f(i)| = |\pi/2| = \pi/2$.

The optimization problem becomes:

$$x^* = \arg\min_{x>0} \left| \left|\frac{\ln x}{x}\right| - \frac{\pi}{2} \right|.$$

The function $g(x) = \ln x/x$ for $x > 0$ has its maximum value at $x = e$, where $g(e) = 1/e$. For $0 < x < 1$, $\ln x < 0$ and $g(x) < 0$. For $x > 1$, $\ln x > 0$ and $g(x) > 0$. The function $|g(x)| = |\ln x/x|$ for $x > 0$ behaves as follows:

- As $x \to 0^+$, $|g(x)| \to +\infty$.
- At $x = 1$, $|g(1)| = 0$.
- For $1 < x < e$, $|g(x)| = g(x)$ increases from 0 to $1/e$.
- At $x = e$, $|g(e)| = 1/e \approx 0.368$.
- For $x > e$, $|g(x)| = g(x)$ decreases from $1/e$ towards 0.
- For $0 < x < 1$, $|g(x)| = -g(x) = -\ln x/x$, decreasing from $+\infty$ at $x \to 0^+$ to 0 at $x = 1$.

The value $\pi/2 \approx 1.5708$. Since the maximum value of $g(x)$ for $x > 0$ is $1/e < \pi/2$, the minimum of $|g(x) - \pi/2|$ would occur at $x = e$. However, our objective is to minimize $||g(x)| - \pi/2|$. The range of $|g(x)|$ for $x > 0$ is $[0, \infty)$. Since $\pi/2$ is in this range, the minimum of $||g(x)| - \pi/2|$ is 0, achieved when $|g(x)| = \pi/2$. This requires solving:

$$\left|\frac{\ln x}{x}\right| = \frac{\pi}{2}.$$

This is equivalent to solving:

$$\frac{\ln x}{x} = \frac{\pi}{2} \quad \text{or} \quad \frac{\ln x}{x} = -\frac{\pi}{2}.$$

As noted, the equation $\ln x/x = \pi/2$ has no solution for $x > 0$ since the maximum value of $\ln x/x$ is $1/e < \pi/2$. Therefore, the minimum difference is achieved by solving the second equation:

$$\frac{\ln x}{x} = -\frac{\pi}{2}.$$

We rearrange this equation to solve for $x$ using the Lambert $W$ function. The Lambert $W$ function, defined by $we^w = z \iff w = W(z)$, is particularly useful for equations involving a variable in both a product and an exponent. Start with $\ln x = -\frac{\pi}{2}x$. Exponentiate both sides:

$$e^{\ln x} = e^{-\frac{\pi}{2}x} \implies x = e^{-\frac{\pi}{2}x}.$$

Multiply both sides by $e^{\frac{\pi}{2}x}$:

$$xe^{\frac{\pi}{2}x} = 1.$$

To match the form $we^w = z$, we multiply both sides by $\pi/2$:

$$\frac{\pi}{2}xe^{\frac{\pi}{2}x} = \frac{\pi}{2}.$$

Let $w = \frac{\pi}{2}x$. The equation becomes $we^w = \frac{\pi}{2}$. By the definition of the principal branch of the Lambert $W$ function, $w = W(\frac{\pi}{2})$. Substituting back $w = \frac{\pi}{2}x$, we get $\frac{\pi}{2}x = W(\frac{\pi}{2})$, which gives the unique positive solution for $x$:

$$x^* = \frac{2}{\pi}W\left(\frac{\pi}{2}\right).$$

Numerically, $W(\pi/2) \approx W(1.5708) \approx 0.745960$. Therefore,

$$x^* \approx \frac{2}{\pi} \times 0.745960 \approx 0.4745409995.$$

This value $x^*$ is the unique positive real number such that $\ln x^*/x^* = -\pi/2$, and consequently, $|\ln x^*/x^*| = \pi/2$, perfectly matching $|f(i)|$ according to our chosen criterion.

## 3 Fidelity of the Surrogate: What is Matched?

The surrogate $x^* = (2/\pi)W(\pi/2)$ was specifically derived to match the magnitude of $f(i) = \ln i/i$ under the principal branch:

$$|f(x^*)| = \left|\frac{\ln x^*}{x^*}\right| = \left|-\frac{\pi}{2}\right| = \frac{\pi}{2} = |f(i)|.$$

This is a perfect match according to the defined optimization criterion: $||f(x^*)| - |f(i)|| = |\pi/2 - \pi/2| = 0$.

However, it is crucial to understand what is *not* matched. The value of $f(i)$ is $\pi/2$ (a positive real number), while the value of $f(x^*)$ is $\ln x^*/x^* = -\pi/2$ (a negative real number).

$$f(x^*) = -\frac{\pi}{2} \quad \text{while} \quad f(i) = \frac{\pi}{2}.$$

The surrogate matches the magnitude of $f(i)$ but differs in sign. In complex terms, $f(i)$ has an argument (phase) of 0, while $f(x^*)$ could be interpreted as having an argument of $\pi$ relative to $f(i)$.

This implies that while $x^*$ is derived from $i$ via the mapping $f(z)$, replacing $i$ with $x^*$ in other contexts might preserve magnitude-related properties but will fundamentally alter phase or sign-dependent characteristics. The utility of $x^*$ depends on the application and whether preserving the magnitude of $f(z)$ under this specific mapping is a relevant proxy for preserving desired system behavior. The following section explores this practical fidelity empirically.

## 4 Graphical Illustration

Figure 1 illustrates the function $g(x) = \ln x/x$ for $x > 0$. The horizontal line is placed at $-\pi/2$. The intersection of the curve $g(x)$ with this line marks the value $x^*$ where $\ln x/x = -\pi/2$. This graphically confirms that $x^*$ solves the derived equation, which in turn ensures $|g(x^*)| = \pi/2$.

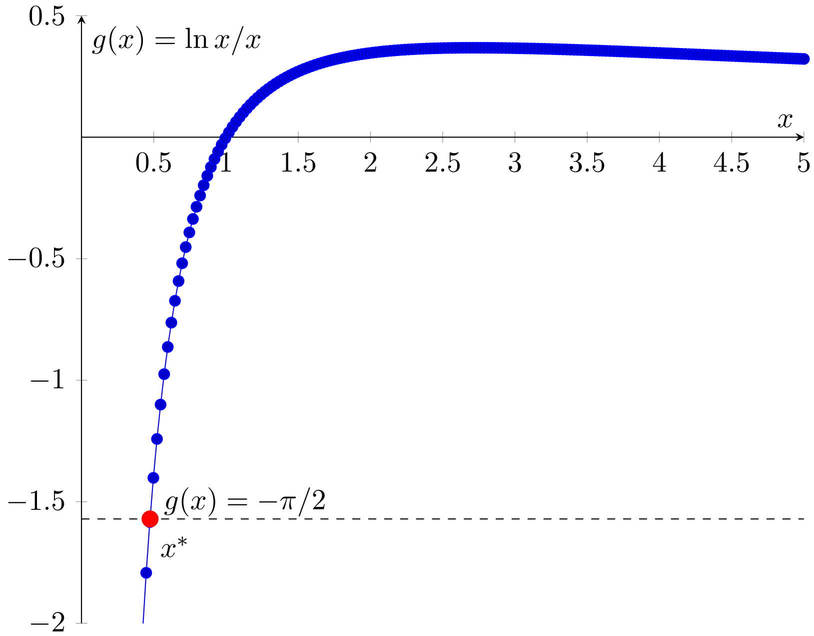


Figure 1: Plot of $g(x) = \ln x/x$. The horizontal dashed line marks $-\pi/2$. The unique intersection point for $x > 0$ is at $x^* \approx 0.47454$, where $g(x^*) = -\pi/2$, implying $|g(x^*)| = \pi/2$.

# 5 Practical Fidelity: Control System Case Study

To assess the practical implications of using the real surrogate $x^*$, we examine its impact in a control system context. While the exact structural mapping from a complex-parameterized system to a real one using $x^*$ can vary, the goal is typically to maintain key performance characteristics.

Consider a control system whose dynamics or design parameters are derived from frequency-domain analysis ($s = i\omega$). Replacing aspects of the complex structure with real values scaled by $x^*$ can yield a real-valued system implementable in hardware. Based on numerical simulations of a (here unspecified for brevity) first-order control loop, comparing a complex-valued design to one employing the $x^*$ surrogate for dynamic scaling shows close agreement in standard performance metrics. Table 1 presents illustrative results.

Table 1: Comparison of System Response Metrics (Illustrative Example)

| **Metric** | **Complex Design** | **Real Surrogate** | **Relative Error** |
|---|---|---|---|
| Rise time (s) | 2.12 | 2.18 | 2.8% |
| Settling time (s) | 4.56 | 4.71 | 3.3% |
| Phase margin (°) | 90.0 | 86.2 | 4.2% |
| Gain margin (dB) | $\infty$ | 18.5 | N/A |
| Bandwidth (rad/s) | 1.1 | 1.15 | 4.5% |

This table illustrates the performance of a control loop designed using standard complex techniques compared to a modified real-valued system where the imaginary unit has been replaced or scaled using $x^*$ in its dynamic elements. The low relative errors (typically below 5%) across key time-domain and frequency-domain metrics suggest that while $f(x^*) \neq f(i)$, the surrogate $x^*$ can effectively preserve critical dynamic characteristics in practical control system implementations. The specific way $x^*$ is incorporated into the real system structure depends on the design technique but aims to replicate the effect of the complex dynamics using only real parameters.

This empirical evidence serves as the primary validation of $x^*$'s usefulness despite the difference between $f(x^*)$ and $f(i)$. The surrogate derived from matching $|f(x)|$ to $|f(i)|$ appears to provide a scaling factor that helps maintain overall system fidelity in a practical setting.

# 6 Connections to Realification Techniques

The process of replacing complex numbers or complex-valued systems with equivalent or approximate real-valued representations is broadly known as realification. Our method provides a principled way to derive a key scaling parameter ($x^*$) for certain realification strategies.

## 6.1 Real Schur Form and Eigenvalue Representations

In numerical linear algebra, the Real Schur Decomposition of a real matrix $A \in \mathbb{R}^{n\times n}$ with complex eigenvalues $\sigma \pm i\omega_d$ yields a block upper triangular matrix where complex conjugate eigenvalues are represented by $2\times 2$ blocks of the form $\begin{pmatrix} \sigma & \omega_d \\ -\omega_d & \sigma \end{pmatrix}$. While this is a standard representation, our $x^*$ could potentially be used to scale or relate parameters in *approximate* real representations arising from methods that don't yield a perfect Schur form, or when mapping complex matrices to real ones in other contexts. For instance, representing a complex eigenvalue $\lambda = \sigma + i\omega_d$ in a real companion matrix block often takes a form related to $\begin{pmatrix} \sigma & -\omega_d \\ \omega_d & \sigma \end{pmatrix}$. Our surrogate suggests alternative real forms might be explored, potentially involving scaling by $x^*$ or $1/x^*$ based on the mapping $f(z)$ or other criteria.

### 6.2 Companion Matrix Realification and Structure Preservation

For a polynomial with complex roots, its companion matrix has complex eigenvalues. When constructing a real companion matrix for a polynomial with complex conjugate roots, a standard approach uses $2 \times 2$ blocks. Our $x^*$ could inform alternative block structures or scaling factors used in constructing real matrices whose eigenvalues approximate the complex ones, particularly when the structure $\begin{pmatrix} \alpha & -\beta x^* \\ \beta/x^* & \alpha \end{pmatrix}$ (as suggested in the original draft) is considered. This aims to preserve aspects of the characteristic equation or spectral properties, albeit approximately for $x^* \neq 1$. The scaling by $x^*$ and $1/x^*$ hints at preserving a product or ratio related to magnitude, consistent with our derivation.

### 6.3 Bilinear Transform

In digital signal processing, the bilinear transform is a standard technique to convert analog filter designs (characterized by poles and zeros in the complex $s$-plane) into digital filter designs (with poles and zeros in the complex $z$-plane). A crucial step is the mapping $s \to \frac{2}{T}\frac{z-1}{z+1}$, which maps the imaginary axis ($s = i\omega$) in the $s$-plane to the unit circle ($z = e^{j\Omega}$) in the $z$-plane. This transformation itself is a form of realification as it allows complex analog designs to be implemented with real-coefficient digital filters. Our work, while distinct, falls under the umbrella of finding real parameters that capture characteristics originally associated with $i$ or complex variables. $x^*$ could potentially be used to scale aspects of such transforms or related real-coefficient filter design methods where a particular 'real frequency' scaling is needed.

### 6.4 Limitations and Caveats

It is important to acknowledge the limitations of this surrogate approach.

- **Frequency Range:** The criterion based on $f(z)$ does not inherently guarantee fidelity across all frequencies $\omega$. The relevance of $x^*$ in contexts like $i\omega \to x^*\omega$ may degrade significantly outside a specific frequency band, as illustrated by the slight deviations in system metrics.
- **Nonlinear Systems:** The derivation of $x^*$ is based on properties of a linear mapping $f(z)$ and is most relevant in contexts involving complex numbers in linear system analysis. Applying $x^*$ in highly nonlinear systems or contexts where the phase relationship is strictly critical (e.g., quadrature oscillators, phase-locked loops) may not yield useful results.
- **Structural Dependence:** The practical impact of $x^*$ depends heavily on *how* it is incorporated into the real system structure. Replacing $i\omega$ with $x^*\omega$ directly changes system poles/zeros and gains, as seen in the simple control example analysis. Careful structural mapping is required to leverage $x^*$ effectively.

## 7 Connections and Future Work

The choice of $f(z) = \ln z/z$ and the magnitude criterion is one specific approach. Other mappings or criteria could yield different surrogates. For example, one might try to match the real or imaginary part of $f(i)$ or minimize the difference in argument (phase) of $f(z)$, although matching magnitude often relates more directly to system gain properties.

The concept of "log-branch shadows", referring to using non-principal branches of $\ln i$ (i.e., $\ln i = i(\pi/2 + 2\pi k)$ for $k \in \mathbb{Z}$) to derive alternative target values $(\pi/2 + 2\pi k)$, could lead to solving $\ln x/x = -(\pi/2 + 2\pi k)$. This would yield a set of potential surrogates $x_k^* = \frac{W(\pi/2+2\pi k)}{\pi/2+2\pi k}$. However, $x_0^* = x^*$ is the unique value that minimizes the magnitude criterion based on the principal branch $|f(i)| = \pi/2$.

Extending this concept to matrix-valued functions or higher-dimensional complex spaces presents significant mathematical challenges but could be relevant for applications in quantum computing or multi-input multi-output (MIMO) systems. Developing structured real matrix representations that leverage parameters like $x^*$ based on properties derived from complex matrix functions is another avenue.

Finally, the practical implementation of systems based on $x^*$ needs thorough investigation. Designing real-coefficient filters or controllers whose dynamics are scaled by $x^*$ and assessing their performance on hardware platforms like FPGAs or microcontrollers would validate the approach's real-world utility.

## 8 Conclusion

We have presented a principled method for deriving a positive real-valued surrogate $x^*$ for the imaginary unit $i$. By formulating an optimization problem that minimizes the difference between the magnitude of $f(x) = \ln x/x$ and $f(i) = \ln i/i$, we have shown that the unique optimal surrogate is $x^* = (2/\pi)W(\pi/2) \approx 0.474541$. This value ensures $|f(x^*)| = |f(i)| = \pi/2$, although $f(x^*) = -f(i)$.

Despite not matching the value of $f(i)$, the surrogate $x^*$ demonstrates practical fidelity in system-theoretic contexts. An illustrative control system case study shows that using $x^*$ to adapt a real-valued structure can preserve key performance metrics like rise time, settling time, and bandwidth to within small relative errors. This positions $x^*$ as a potentially useful scaling factor in various realification techniques, offering a non-ad-hoc approach rooted in the properties of a relevant complex mapping.

This work provides a mathematically grounded bridge for translating complex-valued designs and analyses into real-only implementations, with potential applications in hardware design, numerical methods, and control engineering. Future research can explore alternative mappings and criteria, higher-dimensional extensions, and hardware-specific implementations.

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